Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Founding AI engineer with 8+ years building production systems and the full agentic AI stack end-to-end: pipeline architecture, HITL review workflows, and eval/monitoring infrastructure. Built both halves of what modern AI teams need — **Leasel** (8-agent LangGraph pipeline, conditional HITL escalation when confidence below threshold, 93.6% precision on 1,240 enterprise compliance documents) and **AgentProbe** (3-agent LangGraph runtime with 5-metric LLM-as-judge evaluation engine, CI-integrated regression gating, RAG observability — live at github.com/venkatalakshmiparimala/AgentProbe). Shipped Entrata's first production AI pipeline across 1.6M financial records with compliance controls; 8 years of regulated-data production engineering. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Agentic Workflow Architecture & HITL Design (LangGraph, LangChain) | LLM Evaluation & Observability (LLM-as-Judge, CI Regression Gating) | RAG Pipelines & Vector Search (ChromaDB) | Python · SQL · TypeScript · FastAPI | AWS (S3, EC2, RDS, Glue) · Docker · GitHub Actions | Enterprise Data Pipelines & ETL (1.6M records, compliance gating) | Multi-Provider LLM Routing (Anthropic, OpenAI, Groq, Ollama) | Full-Stack AI Systems (React, Node.js, PostgreSQL)

WORK EXPERIENCE

Magizhs Tech Solution

May 2025 – Aug 2025

AI/ML Engineer Intern

- **Designed and deployed an agentic mission-planning system** as a LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing) — operationalized across marketing and commercial survey workflows, delivering risk-aware compliance decisions with grounded, cited reasons.
- **Built a RAG retrieval layer** (ChromaDB vector engine, FAA Part 107 corpus) grounding every compliance check in source regulation — eliminating AI hallucination on domain-specific rules. Engineered validation guardrails blocking non-compliant outputs before export.

Entrata — PropTech / Real Estate SaaS

Sep 2018 – Aug 2024

Senior Software Engineer — Lehi, UT (Remote)

- **Shipped Entrata's first production AI pipeline** across 1.6M financial records (Python, PyTorch, OpenAI API) — LLM-powered automation with output validation, quality gates, and compliance controls; built streaming data workflows replacing manual Excel operations at enterprise scale.
- **Built enterprise data pipelines and ETL workflows** on AWS (S3, Glue, RDS, EC2) for real-time financial operations (HAP, HUD payments, income eligibility, AMI-based pricing) across 100K+ unit portfolios — optimized 200+ SQL queries (30% performance gain) via execution plan tuning, indexing, and partitioning; 99.5% uptime over 6+ months.
- **Delivered 20+ full-stack production modules** across 120+ Agile sprints for 100K+ monthly active users — cross-functional collaboration with compliance, operations, and product teams; zero federal/state compliance violations.
- **Achieved 100% PHI compliance** on AWS S3 migration with Libsodium-encrypted pipelines — zero security incidents. Led Docker-based microservices redesign: 50% scalability improvement, 30% deployment time reduction.
- Eliminated 80% of manual operational workload via Python/Bash automation; conducted 300+ peer code reviews; mentored 5+ engineers.

George Mason University

Aug 2024 – Aug 2025

Research Associate / Graduate Teaching Assistant

- Built multi-source prediction models (Python, R) achieving $R^2 = 0.977$ across 190+ countries; applied Isolation Forest anomaly detection across 3 research pipelines; designed faculty dashboards improving comprehension 40%. Mentored 600+ students in Python and database design.

PROJECTS

AgentProbe: Multi-Agent Evaluation & LLM Observability Platform Live

3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama), tool-use patterns, and inter-agent state handoffs. CI-integrated 5-metric LLM-as-judge evaluation engine (factual accuracy, hallucination, retrieval, completeness, compliance) — every commit auto-scored, regressions blocked before production. ChromaDB RAG pipeline with trace-level observability: per-run metrics, regression drift, and retrieval quality monitoring.

Python · LangGraph · LangChain · FastAPI · ChromaDB · React · Docker · GitHub Actions | github.com/venkatalakshmiparimala/AgentProbe

Leasel: Enterprise AI Document Intelligence & 8-Agent HITL Pipeline

Three-surface enterprise AI knowledge platform for compliance operations: (1) Document Intelligence — OCR ingestion, field extraction, 100% citation correctness, document diff, missing/conflicting clause detection, human-review gate on regulated decisions; (2) Support Investigation — 8-agent LangGraph pipeline producing ranked, evidence-backed root-cause hypotheses across business rules, settings, and historical incidents; confidence-gated HITL escalation when score drops below threshold; (3) Self-Service Routing — document/policy questions answered directly; eligibility/configuration always routed to staff, never resolved by AI. Core rule enforced in code: LLM extracts and reasons; human or deterministic logic makes the final call. 93.6% precision / 87.9% F1 across 10 compliance rules on 1,240 enterprise documents (5 US states + federal).

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

AI on FHIR: NLP-Driven Medical Records Query System

Reduced manual clinical retrieval time ~60% by building an NLP platform (LLMs, BERT, embeddings) translating natural language queries (EN/ES) to FHIR APIs with HIPAA-compliant RBAC. Deployed on cloud infrastructure with Docker containerization.

Flask · React · LLMs · FHIR · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017

SKILLS

Languages: Python, SQL, TypeScript, JavaScript, PHP, R, Bash

AI & Agents: LangGraph, LangChain, Multi-Agent Orchestration, HITL Design, LLM Evaluation, LLM-as-Judge, Prompt Engineering, RAG Pipelines, Vector Search, ChromaDB, Embeddings, OpenAI API, Anthropic API, Groq, Ollama, FastAPI

ML & Data: PyTorch, Scikit-learn, XGBoost, Isolation Forest, Feature Engineering, NumPy, Pandas, spaCy, BERT, Hugging Face, Tableau

Cloud & Infra: AWS (S3, EC2, RDS, Glue), Azure, Docker, GitHub Actions CI/CD, PostgreSQL, MySQL, Redis, ETL, Data Pipelines, Microservices, RabbitMQ, Node.js